Vulnerability Assessment Report

1st January 2025

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2025 to August 2025. NIST_SP.pdf is used to guide the risk analysis of the information system.

Purpose

The database server is crucial to the business as it stores large amounts of customer, campaign, and analytic data used by employees for querying. Securing the server is crucial as it contains confidential business information that can be stolen. If the database was disabled, customers would lose trust in the company, employees may find it difficult to do their jobs, and the company could face significant public and revenue damage. It is critical to secure the system to maintain integrity and protect business operations.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Employee	*Altering and/or Deleting Information*	*2*	*3*	*6*
Competitor	*Distributed Denial of Service Attack*	*2*	*3*	*6*
Hacker	*Gaining Access to Sensitive Data*	*3*	*3*	*9*

Approach

I selected employees, competitors, and hackers for my risk assessment as they are the most common threat sources organizations encounter. Employees working with the database server on a daily basis could easily alter or delete information whether accidental or not. Competitors may find it easy to access the exposed server and may take advantage in order to advance their own business. Hackers may be financially motivated to steal data off the server. I evaluated each threat by estimating the likelihood and severity then identified the risk for each by multiplying both. This approach allowed me to focus on these crucial threats as they pose a significant business risk by directly impacting confidentiality, integrity, and availability.

Remediation Strategy

To prevent potential data breaches and ensure that only authorized users access the database server, the company should implement least privilege, authentication, and authorization. They should also incorporate multi-factor authentication, encourage the use of strong passwords, and utilize encryption to prevent random users from connecting to the database. Furthermore, the organization could consider implementing backup and recovery procedures to help with remediation in case a data breach does occur.